

```

set enable_hashjoin = off;
explain (analyze, buffers)
with category_stats as (
    select
        p.product_category,
        count(*) as orders_count
    from opt_orders o
    join opt_clients c
        on c.id = o.client_id
    join opt_products p
        on p.product_id = o.product_id
    where c.status = 'active'
        and o.order_date >= date '2023-01-01'
        and o.order_date < date '2024-01-01'
    group by p.product_category
)
select
    (
        select product_category
        from category_stats
        order by orders_count desc, product_category asc
        limit 1
    ) as most_popular_category,
    (
        select product_category
        from category_stats
        order by orders_count asc, product_category asc
        limit 1
    ) as least_popular_category,
    (
        select avg(orders_count)
        from category_stats
    ) as avg_orders_per_category;
reset enable_hashjoin

```

QUERY PLAN

Result (cost=18613.28..18613.29 rows=1 width=268) (actual time=548.311..548.476 rows=1.00 loops=1)

Buffers: shared hit=2669, temp read=541 written=543

CTE category_stats
-> Finalize GroupAggregate (cost=18611.39..18612.90 rows=5 width=18) (actual time=548.185..548.364 rows=5.00 loops=1)
Group Key: p.product_category
Buffers: shared hit=2669, temp read=541 written=543
-> Gather Merge (cost=18611.39..18612.79 rows=12 width=18) (actual time=548.170..548.345 rows=15.00 loops=1)
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=2669, temp read=541 written=543
-> Sort (cost=17611.37..17611.38 rows=5 width=18) (actual time=245.135..245.140 rows=5.00 loops=3)
Sort Key: p.product_category
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=2669, temp read=541 written=543
Worker 0: Sort Method: quicksort Memory: 25kB
Worker 1: Sort Method: quicksort Memory: 25kB
-> Partial HashAggregate (cost=17611.26..17611.31 rows=5 width=18) (actual time=244.886..244.892 rows=5.00 loops=3)
Group Key: p.product_category
Batches: 1 Memory Usage: 32kB
Buffers: shared hit=2653, temp read=541 written=543
Worker 0: Batches: 1 Memory Usage: 32kB
Worker 1: Batches: 1 Memory Usage: 32kB
-> Nested Loop (cost=13435.28..17405.64 rows=41124 width=10) (actual time=110.779..224.035 rows=33289.00 loops=3)
Buffers: shared hit=2653, temp read=541 written=543

-> Merge Join (cost=13434.99..16348.56 rows=41124 width=4) (actual time=110.451..184.915 rows=33289.00 loops=3)
Merge Cond: (o.client_id = c.id)
Buffers: shared hit=1751, temp read=541 written=543
-> Sort (cost=13424.02..13629.57 rows=82220 width=20) (actual time=110.003..130.633 rows=66567.00 loops=3)
Sort Key: o.client_id
Sort Method: external merge Disk: 4328kB
Buffers: shared hit=999, temp read=541 written=543
Worker 0: Sort Method: quicksort Memory: 1688kB
Worker 1: Sort Method: quicksort Memory: 1908kB
-> Parallel Index Only Scan using idx_opt_orders_date_client_product on opt_orders o (cost=0.42..6711.90 rows=82220 width=20) (actual time=0.367..39.143 rows=66568.33 loops=3)
Index Cond: ((order_date >= '2023-01-01'::date) AND (order_date < '2024-01-01'::date))
Heap Fetches: 0
Index Searches: 1
Buffers: shared hit=991
-> Index Only Scan using idx_opt_clients_status_id on opt_clients c (cost=0.42..1987.72 rows=50017 width=16) (actual time=0.430..19.510 rows=50024.00 loops=3)
Index Cond: (status = 'active'::text)
Heap Fetches: 0
Index Searches: 3
Buffers: shared hit=752
-> Memoize (cost=0.29..0.30 rows=1 width=14) (actual time=0.001..0.001 rows=1.00 loops=99867)
Cache Key: o.product_id

Cache Mode: logical
Hits: 73360 Misses: 100 Evictions: 0 Overflows: 0 Memory Usage: 12kB
Buffers: shared hit=902
Worker 0: Hits: 11678 Misses: 100 Evictions: 0 Overflows: 0 Memory Usage: 12kB
Worker 1: Hits: 14529 Misses: 100 Evictions: 0 Overflows: 0 Memory Usage: 12kB
-> Index Scan using opt_products_pkey on opt_products p (cost=0.28..0.29 rows=1 width=14) (actual time=0.009..0.009 rows=1.00 loops=300)
Index Cond: (product_id = o.product_id)
Index Searches: 300
Buffers: shared hit=902
InitPlan 2
-> Limit (cost=0.13..0.13 rows=1 width=126) (actual time=548.238..548.240 rows=1.00 loops=1)
Buffers: shared hit=1279, temp read=541 written=543
-> Sort (cost=0.13..0.14 rows=5 width=126) (actual time=548.236..548.237 rows=1.00 loops=1)
Sort Key: category_stats.orders_count DESC, category_stats.product_category
Sort Method: top-N heapsort Memory: 25kB
Buffers: shared hit=1279, temp read=541 written=543
-> CTE Scan on category_stats (cost=0.00..0.10 rows=5 width=126) (actual time=548.191..548.214 rows=5.00 loops=1)
Storage: Memory Maximum Storage: 17kB
Buffers: shared hit=1279, temp read=541 written=543
InitPlan 3
-> Limit (cost=0.13..0.13 rows=1 width=126) (actual time=0.027..0.028 rows=1.00 loops=1)
-> Sort (cost=0.13..0.14 rows=5 width=126) (actual time=0.026..0.026 rows=1.00 loops=1)

Sort Key: category_stats_1.orders_count, category_stats_1.product_category
Sort Method: top-N heapsort Memory: 25kB
-> CTE Scan on category_stats category_stats_1 (cost=0.00..0.10 rows=5 width=126) (actual time=0.001..0.002 rows=5.00 loops=1)
Storage: Memory Maximum Storage: 17kB
InitPlan 4
-> Aggregate (cost=0.12..0.12 rows=1 width=32) (actual time=0.035..0.036 rows=1.00 loops=1)
-> CTE Scan on category_stats category_stats_2 (cost=0.00..0.10 rows=5 width=8) (actual time=0.000..0.002 rows=5.00 loops=1)
Storage: Memory Maximum Storage: 17kB
Planning:
Buffers: shared hit=11
Planning Time: 4.659 ms
Execution Time: 586.805 ms